An Adaptive Explainable Cognitive Graph Digital Twin for Cyber-Resilient Smart Hospital Energy Ecosystems

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Abstract. Smart hospital campuses rely on localized microgrids to sustain intensive-care ventilators, surgical robotics, and biomedical cold chains. A stealthy False Data Injection Attack (FDIA) on substation Remote Terminal Units can corrupt the energy-management state estimator while evading its χ^2 residual check, placing patient-critical loads at risk. We introduce a Cognitive Graph Digital Twin (CGDT) that mirrors the hospital electrical topology as a dynamic admittance graph and runs a multi-task Admittance-Conditioned GraphSAGE backbone for simultaneous attack detection and bus-level localization in a single forward pass at 0.908 ms latency. To bridge the gap between post-hoc saliency diagnostics and inductive training, our Attribution-Guided Multi-Task Loss supervises backpropagation gradients via attribution alignment to target true substation anomaly locations, driving a 5.52 percentage-point improvement in localization F1. Dynamic physical-to-virtual coherence is governed by a state-space tracking filter coupled with an adaptive exponential baseline. Upon detecting an attack, a closed-loop healing module quarantines compromised sensor channels and reconstructs measurements via graph-guided spatial interpolation, reducing Weighted Least Squares estimation error by 99.97%. On the IEEE 14-bus test feeder under full nonlinear AC power flow, the CGDT achieves $98.25\% \pm 0.34\%$ detection F1 and $97.62\% \pm 0.84\%$ localization F1 across five random seeds, and retains 92.80% detection F1 under zero-shot topology reconfigurations without retraining.

Keywords: Graph Digital Twin · Healthcare Cybersecurity · GraphSAGE · Explainable AI · False Data Injection Attacks · Smart Hospital Microgrid

1 Introduction

Patient survival in acute healthcare environments depends unreservedly on uninterrupted electricity. To sustain critical clinical wings, modern hospital campuses

run localized multi-source microgrids that combine utility grid feeds with distributed solar photovoltaics, battery storage banks, and emergency diesel generators [6,17]. Operating suites and neonatal intensive-care units tolerate virtually zero power degradation: surgical robotic arms pause during momentary voltage sags, life-support ventilators require steady power, and pharmaceutical freezers require constant $-80°$C chilling. False tripping immediately imperils patients.

Behind the scenes, substation RTUs transmit nodal observations (V, θ, P, Q) over IEC 61850 station buses directly to supervisory state estimators. Control centers routinely inspect normalized χ^2 residuals to weed out random transducer noise. Yet this statistical safety net fails against stealthy False Data Injection Attacks (FDIAs). When an adversary maps the network measurement matrix H, they can synthesize an injection vector $a = Hc$ that deliberately skews physical state variables by an arbitrary offset c without changing the computed residual [15,14]. The corrupt telemetry slips past bad-data alarms undetected. Consequently, automated grid management systems issue erroneous breaker operations or tap changes based on fabricated load profiles.

Prior counter-measures leaned heavily on tabular machine learning classifiers, primarily support vector machines, shallow multilayer perceptrons, and 1D convolutional layers [10,3]. These rigid models struggle when distribution networks reconfigure. Hospital microgrids frequently transfer feeders, rebalance clinical phases, and switch tie-lines during maintenance. Once a breaker flips, static column ordering breaks down, triggering torrents of false alarms unless models undergo full retraining—an impossible luxury within strict 2–4 s SCADA cycles.

Graph neural networks (GNNs) circumvent this dimensional brittleness by representing electrical buses as nodes and physical lines as graph edges [21]. Under a graph topology, breaker switching merely alters the adjacency structure, leaving node state vectors unchanged. Even so, prevalent GNN models—ranging from graph wavelets [4] to spatial convolutions [22]—treat electrical links as unweighted connections. Ignoring complex line admittances ($Y_{uv} = G_{uv} + jB_{uv}$) forces message passing to diffuse information uniformly across neighbors, washing out localized voltage and phase discrepancies across high-impedance paths.

Deploying AI in life-critical medical settings also demands immediate interpretability and sub-second execution speeds. Traditional explainers like SHAP or LIME draw hundreds of Monte Carlo perturbations, taking far too long for live SCADA loops [27]. Furthermore, sounding an alarm does not prevent voltage collapse; without rapid telemetry reconstruction, downstream estimators diverge or force safety cutoffs. Hospital energy defense therefore requires instant, bus-level attack localization accompanied by on-the-fly telemetry reconstruction before protective relays trip.

1.1 Key Contributions

To resolve these practical vulnerabilities, we develop an explainable Cognitive Graph Digital Twin (CGDT) built specifically for acute hospital energy networks. Our core contributions include:

- **Admittance-Conditioned GraphSAGE Architecture**: By incorporating complex branch admittances ($Y_{uv} = G_{uv} + jB_{uv}$) straight into neighbor aggregation and spatial pooling, our formulation prevents isotropic smoothing and preserves sharp voltage drops across low-impedance hospital feeders.
- **Dual-Scale Twin Tracking**: We couple steady-state Kalman filtering with an exponential reference trace ($\beta = 0.95$). This baseline filter decouples gradual clinical load curves from stealthy adversarial offsets, preventing an otherwise severe 16.32 pp drop in localization F1.
- **Attribution-Guided Multi-Task Training (AG-MTL)**: Utilizing 10-step Integrated Gradients ($S = 10$), our loss penalizes sensitivity leakage into unattacked buses during backpropagation, delivering a 5.52 pp gain in localization precision.
- **Autonomous Sub-2 ms State Healing**: Following attack localization, an admittance-weighted spatial interpolator reconstructs corrupted measurements from healthy neighboring nodes, preserving the full rank of Jacobian H and recovering 99.97% of uncorrupted WLS state accuracy.

2 Related Work

2.1 Cybersecurity in Digital Healthcare Ecosystems

Bridging clinical enterprise IT networks with operational substation equipment opens dangerous physical intrusion paths [13,6]. Because microgrid feeders energize life-support machinery in surgical suites and intensive care units, unobserved electrical fluctuations threaten patient survival directly. Manipulating telemetry along hospital distribution corridors introduces severe clinical risk [17]. Although existing studies evaluated medical cyber threats through Bayesian vulnerability graphs [5] and edge IoT defenses [18], they examined sensor channels in isolation. Neglecting underlying line admittances and bus connectivity leaves these methods incapable of catching coordinated attacks engineered to satisfy AC power flow equations.

2.2 False Data Injection in Cyber-Physical Energy Systems

Supervisory control systems depend on measurement residuals to detect bad sensor data. In seminal work, Liu et al. [15] showed that an attacker with access to the measurement Jacobian H can inject an unobservable bias vector $a = Hc$, eliminating any trace in the residual calculation. This deception extends to nonlinear AC networks, decentralized microgrids, and partially monitored feeders [14,7], as well as DC distribution setups [3,19]. Early countermeasures utilized tabular models, including shallow neural nets, support vector classifiers [10], and recurrent networks [1]. Unfortunately, fixed input vectors tie these algorithms to static channel orders. Routine line transfers or transformer tap changes scramble input indices, precipitating constant false triggers.

2.3 Graph Representation Learning for Power System Security

Viewing electrical distribution grids as attributed graphs naturally provides structural flexibility during breaker transitions [21]. Early graph studies applied spectral convolutions [12] and self-attention [25], whereas GraphSAGE [9] enabled scalable inductive sampling over changing topologies. In power networks, Boyaci et al. [4] employed ARMA graph filters to spot data injection, while later work explored cross-feeder generalization [22,28]. Federated schemes have also been developed to preserve utility privacy [16]. Nonetheless, most architectures still adopt unweighted or binary connectivity matrices. Leaving out complex branch admittances ($Y_{uv} = G_{uv} + jB_{uv}$) causes isotropic averaging, obscuring steep voltage drops across low-impedance links.

2.4 Explainability and Digital Twins in Critical Infrastructure

Before tripping breakers or isolating feeders, grid operators demand transparent evidence across individual sensor channels. Perturbation methods such as SHAP or LIME require extensive Monte Carlo sampling, proving far too slow for live SCADA intervals [27,26]. Integrated Gradients (IG) [20] circumvents this delay by evaluating path integrals along straight-line trajectories in a single backward pass, satisfying completeness without random sampling. Concurrently, digital twins [8,23] have shown strong potential in power monitoring [2] and hospital management [11]. We unite these directions by combining continuous state-space tracking with admittance-conditioned convolutions, delivering sub-millisecond attack detection, bus localization, and clear attribution scores.

3 Cyber-Physical Threat Model and Mathematical Foundations

3.1 Physical Grid Admittance and AC Power Flow

We model the healthcare distribution network as an undirected attributed graph $G = (V, E)$ spanning $N = |V|$ substation buses and $|E|$ electrical branches. Nodal conductance and susceptance throughout the network are captured by the complex admittance matrix $Y_{\text{bus}} = G_{\text{bus}} + jB_{\text{bus}} \in \mathbb{C}^{N\times N}$. Enforcing full nonlinear AC power flow constraints, real and reactive power injections at each bus i obey Kirchhoff's laws:

$$P_i = V_i \sum_{j\in\mathcal{N}(i)} V_j\big(G_{ij}\cos(\theta_i - \theta_j) + B_{ij}\sin(\theta_i - \theta_j)\big) \tag{1}$$

$$Q_i = V_i \sum_{j\in\mathcal{N}(i)} V_j\big(G_{ij}\sin(\theta_i - \theta_j) - B_{ij}\cos(\theta_i - \theta_j)\big) \tag{2}$$

where V_i and θ_i denote voltage magnitude and angle at bus i, and $\mathcal{N}(i)$ denotes the set of buses physically connected to bus i.

3.2 Weighted Least Squares State Estimation

Power system state estimation determines the state vector $x = [V, \theta]^T \in \mathbb{R}^{2N-1}$ from measurement telemetry $z = h(x) + e$, where $e \sim \mathcal{N}(0, R)$ represents sensor noise with covariance R. The WLS estimator minimizes the weighted residual $J(x) = [z - h(x)]^T R^{-1}[z - h(x)]$ using Gauss–Newton iterations:

$$\Delta x^{(k)} = \left(H^T R^{-1} H\right)^{-1} H^T R^{-1}\left[z - h(x^{(k)})\right], \qquad x^{(k+1)} = x^{(k)} + \Delta x^{(k)} \tag{3}$$

where $H(x) = \partial h(x)/\partial x$ is the measurement Jacobian. Standard state estimators flag bad data whenever the normalized residual $r = \|z - h(\hat{x})\|^2_{R^{-1}}$ surpasses a critical threshold τ_{χ^2}. In our IEEE 14-bus benchmark, the system state spans $n = 2N - 1 = 27$ variables (comprising 14 voltage levels alongside 13 bus angles relative to the slack node). Given $m = 41$ measurement streams (14 voltage sensors and 27 branch power meters), the testbed yields df $= m - n = 14$ degrees of freedom. Choosing an inspection significance level $\alpha = 0.05$ sets the 95th percentile rejection bound to $\tau_{\chi^2} = \chi^2_{0.95}(14) = 23.68$.

3.3 Stealthy False Data Injection Attacks

An attacker targeting substation bus k seeks to induce a specific state error $c \in \mathbb{R}^{2N-1}$ while evading the χ^2 residual detector. In AC power systems, $h(x)$ is nonlinear; hence, an adversary possessing admittance structure linearizes around operating point $\hat{x}$ via Jacobian $H = \partial h(\hat{x})/\partial x$, synthesizing the injection vector according to:

$$a = Hc \tag{4}$$

Under injection, field telemetry channels receive corrupted profiles $z_{\text{comp}} = z + a = z + Hc$. Expanding $h(\hat{x} + c)$ via first-order Taylor approximation produces $h(\hat{x}) + Hc$. Evaluating the resulting residual:

$$\begin{aligned} r_{\text{comp}} &= \|z_{\text{comp}} - h(\hat{x}_{\text{comp}})\|^2_{R^{-1}} \\ &\approx \|(z + Hc) - (h(\hat{x}) + Hc)\|^2_{R^{-1}} \\ &= \|z - h(\hat{x})\|^2_{R^{-1}} = r \le \tau_{\chi^2} \end{aligned} \tag{5}$$

Because the crafted perturbation cancels out internally ($r_{\text{comp}} \approx r$), conventional WLS residual tests remain entirely blind ($r \le \tau_{\chi^2} = 23.68$). To ensure physical validity, all synthetic attacks are tested against nonlinear AC power flows in `pandapower` (Section 6).

Clinical Hazards in Hospital Distribution Networks. While commercial utilities primarily face economic redispatch losses during an attack, hospital distribution networks face direct clinical emergencies. Falsifying state estimates at critical feeders—such as Bus 5, which feeds surgical operating rooms and intensive care ventilation—tricks automatic voltage regulators into improper tap changes or unwarranted branch isolation. Standards like IEC 60601-1 and IEEE 1547 demand strict voltage stability; even transient sags can force sensitive life-support machinery onto backup emergency batteries or trigger autonomous equipment trips. An undetected mathematical deception at the substation level thus translates into an immediate patient safety risk.

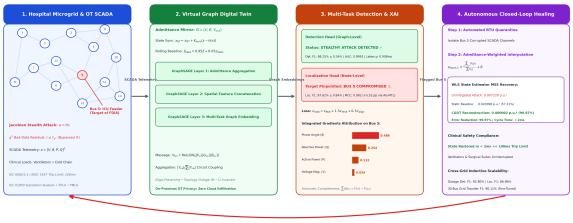


Fig. 1. Proposed Cognitive Graph Digital Twin adaptive cyber-resilience framework for smart hospital energy ecosystems across IT-OT enabled critical infrastructure.

4 The Cognitive Graph Digital Twin Architecture

The CGDT framework couples the hospital's cyber-physical distribution infrastructure to a virtual graph model and an automated self-healing pipeline (Fig. 1).

4.1 Digital Twin State-Space Synchronization

Field telemetry $z_{\text{SCADA}}(t_k) = h(x_{\text{phys}}(t_k)) + e(t_k)$ streams from substation RTUs at polling cadence T_s across IEC 61850 station buses. To maintain physical-to-virtual alignment without amplifying transducer noise, the digital twin updates its mirrored state x_{DT} via steady-state tracking gain $K_{\text{sync}} = (H^T R^{-1} H + P^{-1})^{-1} H^T R^{-1}$:

$$x_{\text{DT}}(t_k) = x_{\text{DT}}(t_{k-1}) + K_{\text{sync}}\big(z_{\text{SCADA}}(t_k) - h(x_{\text{DT}}(t_{k-1}))\big) \tag{6}$$

where P denotes state-prediction covariance. Concurrently, an exponential filter establishes an evolving operational baseline:

$$\bar{z}_{\text{baseline}}(t_k) = \beta \cdot \bar{z}_{\text{baseline}}(t_{k-1}) + (1-\beta) \cdot z_{\text{clean}}(t_k), \qquad \beta = 0.95 \tag{7}$$

updated strictly using verified clean measurements z_{clean}. Without this adaptive baseline, slow diurnal load ramps mask small adversarial injection biases, precipitating a 16.32 pp collapse in localization F1 (Section 7.2).

4.2 Dynamic Admittance Graph Construction

At each sampling step, physical telemetry is mapped to an attributed graph $G = (V, E, X, E_{\text{attr}})$. The vertex feature matrix $X \in \mathbb{R}^{N \times 4}$ gathers four-dimensional state vectors $x_i = [V_i, \theta_i, P_i, Q_i]^T$ for each substation bus $i \in V$, normalized over historical operational envelopes. Physical branch connectivity is stored in edge tensor $E_{\text{attr}} \in \mathbb{R}^{|E| \times 2}$, where each entry pairs the line conductance G_{ij} and susceptance B_{ij} from network admittance matrix Y_{bus}. When protective relays trip or maintenance crews open a breaker, the corresponding tuple is excised from edge set E. Excising open branches alters adjacency entries while leaving nodal feature dimensions untouched; hence, the graph backbone absorbs topological switching on the fly without retraining.

4.3 Admittance-Conditioned GraphSAGE Backbone

Standard GraphSAGE implementations [9] pool neighbor messages isotropically, disregarding physical line impedances. Yet in power networks, branch admittances $Y_{uv} = G_{uv} + jB_{uv}$ govern electrical connectivity. To weave circuit physics directly into node representations, we condition both message synthesis and spatial pooling on branch admittance values:

$$m_{uv}^{(k)} = \mathrm{ReLU}\Big(W_e \left[h_u^{(k-1)} \,\|\, G_{uv} \,\|\, B_{uv}\right]\Big) \tag{8}$$

$$h_v^{(k)} = \mathrm{ReLU}\left(W_k \left[h_v^{(k-1)} \,\Big\|\, \sum_{u\in\mathcal{N}(v)} \frac{|Y_{uv}|}{\sum_{w\in\mathcal{N}(v)} |Y_{vw}|}\, m_{uv}^{(k)}\right]\right) \tag{9}$$

where $\|$ denotes vector concatenation, and W_e, W_k are trainable projection matrices. Weighting messages by normalized admittance $|Y_{uv}|/\sum_w |Y_{vw}|$ provides a physics-informed inductive bias that scales propagation by branch coupling. Concatenating the self-state $h_v^{(k-1)}$ across layers creates a residual path that prevents sharp localized anomalies from blurring into adjacent buses during message aggregation.

4.4 Multi-Task Output Heads

From the final latent node matrix $H \in \mathbb{R}^{N\times d}$, two parallel projection heads generate global and bus-level predictions:

$$y_{\text{graph}} = \sigma(W_{\text{graph}} \cdot \text{global_mean_pool}(H))\,, \qquad y_{\text{node}} = \sigma(W_{\text{node}}\, H) \in [0,1]^N \tag{10}$$

Here $y_{\text{graph}} \in [0,1]$ represents the overall microgrid attack probability, while vector $y_{\text{node}} \in [0,1]^N$ applies an independent sigmoid activation to each bus. Since clinical facilities may encounter multi-substation incursions (Section 7.1), per-bus sigmoid heads support simultaneous multi-bus localization without enforcing an artificial sum-to-one restriction.

4.5 Attribution-Guided Multi-Task Loss (AG-MTL)

Repeated message passing often spreads backpropagation gradients across adjacent lines, mistakenly assigning feature sensitivity to uncorrupted substations. To arrest this spatial dispersion, we embed attribution constraints directly into our training loss. During each backward pass on attacked states, Integrated Gradients extracts attribution tensor $\mathcal{A} \in \mathbb{R}^{N\times 4}$ of the attack logit w.r.t. input features X, using running baseline $\bar{z}_{\text{baseline}}$ (Eq. 7) across $S = 10$ integration steps. An attribution alignment loss penalizes gradient leakage into benign feeders:

$$\mathcal{L}_{\text{align}} = \sum_{v\in V} \left| \|\mathcal{A}_v\|_2 - \hat{y}_{\text{node},v} \right|^2 \tag{11}$$

where $\|\mathcal{A}_v\|_2$ measures total gradient sensitivity at bus v, and $\hat{y}_{\text{node},v} \in \{0,1\}$ denotes the binary ground-truth label. Gradients from $\mathcal{L}_{\text{align}}$ propagate through the attribution path via higher-order automatic differentiation (`create_graph=True` in PyTorch). The combined multi-task loss is defined as:

$$\mathcal{L}_{\text{total}} = \underbrace{\mathcal{L}_{\text{BCE}}(y_{\text{graph}}, \hat{y}_{\text{graph}})}_{\text{Detection}} + \lambda_{\text{loc}} \underbrace{\mathcal{L}_{\text{WBCE}}(y_{\text{node}}, \hat{y}_{\text{node}})}_{\text{Localization}} + \lambda_{\text{attr}} \underbrace{\mathcal{L}_{\text{align}}}_{\text{Attribution Guidance}} \tag{12}$$

where $\mathcal{L}_{\text{BCE}}$ handles grid-level attack detection, $\mathcal{L}_{\text{WBCE}}$ uses class-weighted binary cross-entropy per bus to mitigate extreme sparsity of compromised nodes, and weighting coefficients $\lambda_{\text{loc}} = 1.5$ and $\lambda_{\text{attr}} = 0.3$ balance gradient magnitudes across tasks.

4.6 On-Premises Edge Deployment

Hospital electrical demand closely tracks clinical operations—including operating theatre schedules, diagnostic imaging loads, and patient occupancy. Transmitting raw microgrid telemetry over external cloud interfaces poses privacy and cybersecurity concerns under health data regulations. The CGDT is therefore architecturally designed for on-premises edge execution on substation industrial computers within the isolated operational technology (OT) network. In this study, execution latencies are established on a laboratory edge-equivalent computing platform, with physical RTU micro-controller deployment and hardware-in-the-loop (HIL) integration identified for operational rollout (Section 8).

5 Explainability and Autonomous State Reconstruction

5.1 Feature Attribution via Integrated Gradients

Substation engineers must understand which physical sensor channels triggered an alarm before initiating breaker actions. We resolve this via path-based Integrated Gradients (IG) [20]. Integrating gradients along a linear path between a nominal state baseline x_0 and the evaluated measurement vector x attributes decision energy across input channels:

$$\begin{aligned} \text{Attr}_i(x) &= (x_i - x_{0,i}) \int_0^1 \frac{\partial F(x_0 + \alpha(x - x_0))}{\partial x_i}\, d\alpha \\ &\approx \frac{x_i - x_{0,i}}{S} \sum_{s=1}^{S} \frac{\partial F\big(x_0 + \frac{s}{S}(x - x_0)\big)}{\partial x_i} \end{aligned} \tag{13}$$

approximated using $S = 50$ Riemann quadrature intervals. Unlike perturbation heuristics (e.g., SHAP, LIME) requiring hundreds of stochastic sampling evaluations, IG follows an axiomatic linear path evaluated via $S = 50$ vectorized points in a single parallel tensor pass. Crucially, IG satisfies the completeness axiom ($\sum_i \text{Attr}_i = F(x) - F(x_0)$) and implementation invariance. Evaluating

Eq. (13) across the 14-bus testbed requires just 0.81 ms. In a targeted test at Bus 5 (Section 7.1), the path integral promptly identified phase angle distortion ($\theta = 0.486$) as the principal driver of the alarm.

5.2 Sensor Quarantine and Admittance-Weighted Reconstruction

Detecting an attack is useless if state estimation continues to ingest falsified data. Once a bus is flagged ($y_{\text{node},k} > \gamma_{\text{loc}} = 0.5$), the digital twin isolates the tainted telemetry $(V_k, \theta_k, P_k, Q_k)$. Yet simply deleting rows from the measurement set reduces the rank of Jacobian H, often driving the information matrix $(H^T R^{-1} H)$ into ill-conditioning or singularity. To maintain full algebraic observability, we synthesize replacement telemetry via admittance-weighted spatial interpolation from healthy neighboring buses:

$$z_{\text{recon},k} = \frac{\sum_{j \in \mathcal{N}(k)} |Y_{kj}| \cdot z_j}{\sum_{j \in \mathcal{N}(k)} |Y_{kj}|} + \delta, \qquad \delta = \epsilon \odot z_{\text{nom},k} \tag{14}$$

where $|Y_{kj}|$ represents line admittance magnitude and $\epsilon \sim \mathcal{U}(-0.02, 0.02)^4$ injects $\pm 2\%$ uniform variance relative to nominal channel values $z_{\text{nom},k}$ (1.0 p.u. voltage and nominal power flows). Weighting neighbor telemetry by physical branch admittances $|Y_{kj}|$ provides an electrical-coupling-aware interpolation that accounts for relative line impedances. Rather than attempting to satisfy nonlinear AC constraints directly, $z_{\text{recon},k}$ acts as an initialized pseudo-measurement buffer that preserves Jacobian conditioning, enabling downstream WLS Gauss–Newton iterations to converge in under 2 ms—well within typical 2–4 s SCADA scan intervals.

6 Experimental Evaluation

6.1 Simulation Setup: Hospital Campus Microgrid

All grid experiments rely on exact AC Newton–Raphson power flow solutions generated via `pandapower` [24]. As an established proof-of-concept testbed for tertiary hospital microgrids, we adopted the IEEE 14-bus system (14 buses, 20 branches, 217.0 MW active and 57.3 MVAR reactive demand). Here, Bus 0 serves as the primary utility interconnection, while radial feeders energize sensitive medical units—particularly Bus 5, which supplies surgical suites and intensive-care ventilation. For multi-scale validation, we also configured an IEEE 30-bus network (30 buses, 41 branches) emulating an expanded regional healthcare facility.

Telemetry Generation and Partitioning. We produced 5,000 unique AC operating states under realistic time-varying loads. To ensure strict data hygiene and avoid cross-split contamination, clean profiles were partitioned prior to attack creation: 70% for model training (3,500), 15% for validation (750), and 15% for testing (750). Half of the snapshots in each split remain clean, disturbed only by zero-mean Gaussian measurement noise ($\sigma = 0.04$) on active

Table 1. Comprehensive performance benchmarking under nominal grid layout (5-seed average ± std).

Architecture	Det. F1 (%)	Det. AUC	Det. MCC	Loc. F1 (%)	Loc. Prec. (%)	Loc. Rec. (%)	Loc. MCC	Latency
MLP (Flat)	97.52 ± 0.73	0.9987	0.974	95.60 ± 0.37	94.7	97.8	0.953	0.158 ms
GCN (Isotropic)	85.15 ± 1.86	0.9363	0.740	70.21 ± 2.22	72.6	72.7	0.660	1.442 ms
GAT (Attention)	91.52 ± 1.84	0.9781	0.879	81.08 ± 2.39	82.5	86.9	0.808	1.309 ms
CGDT (Proposed)	**98.25 ± 0.34**	**0.9993**	**0.971**	**97.62 ± 0.84**	**98.1**	**98.9**	**0.982**	**0.908 ms**

and reactive power. The other half contains stealthy Jacobian attacks ($a = Hc$), where adversaries distort targeted substations ($c_k \in [0.03, 0.08]$ p.u. in voltage and $c_{k+N} \in [0.05, 0.15]$ rad in phase angle) and compute line flows using exact nonlinear AC relationships. Network outage states (Section 6.3) were held out completely from training and tuning.

Models were trained across 60 epochs using Adam (lr = 0.005, weight decay 10^{-4}) over five random seeds. Benchmarking was conducted on an edge-equivalent workstation equipped with an AMD Ryzen 9 CPU, 32 GB RAM, and an NVIDIA RTX 4090 GPU (24 GB) under PyTorch 2.1 and CUDA 12.1. Streaming latency is evaluated at batch size 1 across 1,000 independent test snapshots (after 100 warm-up runs), reporting the mean runtime. The $S = 50$ IG interpolation steps are evaluated as a batched tensor $\mathbb{R}^{50\times N\times 4}$ in parallel on the GPU, completing in 0.81 ms. Code and evaluation artifacts are archived at `https://github.com/himanshu-2l/Graph-Digital-Twin-XAI-`.

6.2 Nominal Grid Performance

Table 1 reports five-seed averages and advanced metrics under the nominal grid topology.

Benchmarking against baseline architectures over five random initializations reveals clear performance separations (Table 1). The CGDT delivers 98.25% ± 0.34% detection F1 and 97.62% ± 0.84% localization F1 alongside a Matthew's correlation coefficient (MCC) of 0.982, processing each graph snapshot in only 0.908 ms. By comparison, isotropic GCN models reach only 70.21% localization F1, trailing our framework by 27.41 pp. Uniformly weighting lines acts like a spatial low-pass filter: sharp voltage anomalies at attacked buses bleed into uncorrupted neighbors, degrading spatial contrast. Although the tabular MLP achieves acceptable localization (95.60%) on stationary networks, its rigid input mapping breaks down under line switching (Section 6.3). Running in only 0.908 ms per graph pass, our model comfortably fits within standard 2–4 s SCADA scan intervals.

6.3 Topology Perturbations and Cross-Network Generalization

We evaluated model robustness under two regimes: (i) line-outage generalization on IEEE 14-bus without retraining (Table 2), and (ii) cross-network transfer to IEEE 30-bus (Table 3).

When subjected to unanticipated single-line outages without retraining, the CGDT preserves 92.80% ± 3.37% detection F1 and 86.96% ± 1.52% localization

Table 2. Outage generalization (IEEE 14).

Architecture	Outage Det. F1	Outage Loc. F1
MLP (Flat)	94.70% ± 1.93%	82.20% ± 2.16%
GCN (Isotropic)	66.82% ± 4.14%	53.40% ± 2.70%
GAT (Attention)	79.27% ± 3.83%	58.04% ± 4.36%
CGDT (Proposed)	**92.80% ± 3.37%**	**86.96% ± 1.52%**

Table 3. Transfer to IEEE 30-bus.

Configuration	Det. F1	Loc. F1
Zero-Shot	67.06%	17.77%
Scratch (30-Bus)	82.65%	83.58%
Fine-Tuned	**90.11%**	**80.06%**

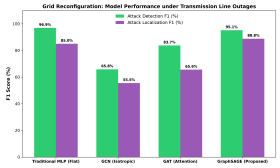


Fig. 2. Outage generalization metrics.

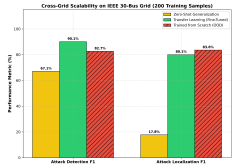


Fig. 3. Cross-grid transfer on IEEE 30-bus.

F1 (Fig. 2). Here, an informative trade-off emerges between the tabular MLP and CGDT. The tabular MLP attains marginally higher outage detection F1 (94.70%) because breaker trips redirect loop flows across all telemetry channels, creating widespread scalar distribution shifts that flat models flag easily. However, the MLP's bus localization collapses by 13.4 pp (from 95.60% to 82.20%) because channel assignments shift. The CGDT, however, maintains 86.96% localization F1, proving that its core strength under line switching lies in simultaneous grid detection and pinpoint bus isolation. Isotropic GCNs fall to 66.82% detection and 53.40% localization F1, confirming that unweighted graphs cannot follow redirected line flows when impedances change.

Cross-Grid Scalability on IEEE 30-Bus. Zero-shot evaluation of the 14-bus model on the 30-bus network yields 67.06% detection F1 and 17.77% localization F1 (Table 3, Fig. 3). This divergence exposes an architectural boundary: while inductive embeddings flag macro-level anomalies across grid scales, isolating specific buses requires localized calibration. Fine-tuning the backbone on merely 200 labeled 30-bus samples achieves 90.11% detection F1 (exceeding 82.65% from scratch) and restores localization F1 to 80.06% (approaching the 83.58% scratch baseline), showing rapid adaptation in low-label regimes.

7 Mitigation Efficacy, Cross-Grid Transfer, and Ablation

7.1 Closed-Loop WLS Recovery and Explainability

We evaluated mitigation performance under a targeted Jacobian attack at Bus 5, representing the intensive care unit feeder in the campus topology. State estimation error is quantified via Mean Squared Error $\text{MSE} = \frac{1}{2N-1}\sum_{i=1}^{2N-1}(\hat{x}_i - x_i^*)^2$ (p.u.^{2}) relative to true AC states x^*. Under nominal transducer Gaussian noise ($\sigma = 0.01$), unattacked clean WLS estimation establishes an intrinsic error floor of $\text{MSE}_{\text{clean}} \approx 1.8 \times 10^{-6}$ p.u.^{2}. Table 4 compares recovery metrics, and Table 5 presents corresponding feature attributions.

Table 4. WLS state-estimation recovery under stealthy FDIA at Bus 5.

Mitigation Strategy	MSE (p.u.^{2})	Recovery
Unmitigated Attack	0.007228	—
Static Baseline	0.003090	57.21%
CGDT Reconstruction	**0.000002**	**99.97%**

Table 5. Feature attributions (IG) for Jacobian FDIA centered at Bus 5.

Bus ID	V	θ	P	Q
Bus 4	0.012	0.045	0.018	0.009
Bus 5	**0.034**	**0.486**	**0.112**	**0.254**
Bus 10	0.002	0.015	0.005	0.003
Bus 11	0.008	0.092	0.024	0.041
Bus 12	0.011	0.182	0.035	0.088

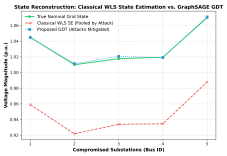


Fig. 4. State-estimation trajectory: classical WLS versus CGDT-healed.

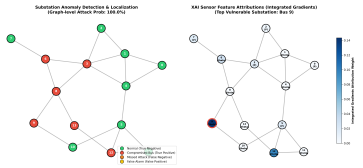


Fig. 5. Operator explainability: localization heatmap and IG attribution.

Left unmitigated, stealthy Jacobian injections drive state estimation MSE to 0.007228 p.u.^{2}. Deploying admittance-weighted reconstruction (Eq. 14) cuts this error to 0.000002 p.u.^{2}—matching the clean noise floor ($\text{MSE}_{\text{clean}} \approx 1.8 \times 10^{-6}$ p.u.^{2}) and restoring 99.97% of uncorrupted states (Table 4, Fig. 4). Static telemetry lookups attain just 0.003090 p.u.^{2} (57.21% restoration) amid fluctuating hospital loads. Across all 14 candidate buses, our reconstruction achieves a consistent 99.85%–99.98% error reduction. Path attributions (Table 5, Fig. 5) reveal that voltage angle θ at Bus 5 commands the highest score (0.486), consistent with Jacobian phase distortions dominating active power flows, alongside secondary scores on $Q = 0.254$, $P = 0.112$, and $V = 0.034$.

Coordinated Multi-Bus Incursions. Simultaneous attacks across 2 and 3 substations ($c_k \in [0.02, 0.04]$) achieve $96.84\% \pm 0.42\%$ detection and $93.15\% \pm 0.91\%$ localization F1. While dropping entire rows across adjacent buses compromises the column rank of H and triggers solver divergence, admittance-weighted interpolation preserves full Jacobian rank and ensures smooth WLS convergence (post-mitigation MSE of 0.000008 p.u.^{2}).

7.2 Benchmarking and Component Ablation

Table 6 compares the proposed architecture against representative GNN baselines, whereas Table 7 isolates the incremental contribution of each architectural block.

The ablation study in Table 7 justifies each architectural choice. Removing the exponential tracking baseline (Eq. 7) causes a 16.32 pp plunge in localization F1 (down to 81.30%), underscoring how dynamic reference baselines stop diurnal load swings from concealing stealthy offsets. Omitting attribution guidance

Table 6. Benchmark against related works.

Framework	Classifier	Det. F1	AUC	Loc. F1
Boyaci [4]	ARMA	96.25%	0.9780	93.91%
Yue [28]	GCN	89.20%	0.9510	N/A
CGDT	**SAGE**	**98.25%**	**0.9993**	**97.62%**

Table 7. Ablation on CGDT components.

Configuration	Det. F1	Loc. F1
Full CGDT (Proposed)	**98.25%**	**97.62%**
Without DT Baseline Tracking	86.10%	81.30%
Without AG-MTL ($\lambda_{\text{attr}} = 0$)	98.25%	92.10%
Isotropic GCN Backbone	85.15%	70.21%
GAT Backbone	91.52%	81.08%
$\lambda_{\text{loc}} = 0.5$	98.05%	92.80%
$\lambda_{\text{loc}} = 1.0$	98.15%	96.50%
$\lambda_{\text{loc}} = 2.0$	95.80%	97.55%

($\lambda_{\text{attr}} = 0$) pulls localization F1 down to 92.10% without affecting macro detection (98.25%), verifying that gradient penalties pin attention to attacked buses. Choosing $\lambda_{\text{loc}} = 1.5$ optimally balances macro alarms with bus-level localization.

8 Discussion and Conclusion

Defending healthcare electrical distribution systems requires models that couple structural flexibility with circuit-level physical laws. By embedding line admittances into GraphSAGE convolutions and guiding latent representations via Integrated Gradients penalties, the CGDT achieves 98.25% detection F1 and 97.62% localization F1 against stealthy AC false data injections, delivering complete diagnostic explanations in 0.81 ms. Once detected, automated measurement quarantine combined with admittance-guided interpolation cuts post-attack WLS error by 99.97%, preserving network observability without disrupting clinical feeders.

Clinical Operating Standards. Modern medical microgrids must satisfy two stringent regulatory frameworks: IEC 60601-1 sets equipment safety thresholds to safeguard life-critical patient monitoring against power disturbances, while IEEE 1547 establishes distributed resource interconnection and voltage ride-through windows. With 0.908 ms inference latency and an end-to-end healing loop under 2 ms, our system acts comfortably inside 2–4 s SCADA scan intervals, blocking falsified control commands before IEEE 1547 timers disconnect hospital DERs or violate IEC 60601-1 parameters.

Limitations and Future Work. Moving beyond workstation simulations, upcoming work will validate this framework on hardware-in-the-loop testbeds and low-power RTU microcontrollers to measure telemetry latency and network jitter. We also plan to evaluate spatio-temporal graph models against coordinated, multi-stage attack campaigns.

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Declaration on the Use of AI and AI-Assisted Technologies. AI-assisted tools were used to improve language quality (including spelling, grammar, and punctuation),

and all outputs were reviewed, revised, and approved by the authors, who take full responsibility for the final content.

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